

1. Number

Structure and calculation

What students need to learn:

- N1** order positive and negative integers, decimals and fractions;
use the symbols $=, \neq, <, >, \leq, \geq$
- N2** apply the four operations, including formal written methods, to integers, decimals and simple fractions (proper and improper), and mixed numbers – all both positive and negative; understand and use place value (e.g. when working with very large or very small numbers, and when calculating with decimals)
- N3** recognise and use relationships between operations, including inverse operations (e.g. cancellation to simplify calculations and expressions); use conventional notation for priority of operations, including brackets, powers, roots and reciprocals
- N4** use the concepts and vocabulary of prime numbers, factors (divisors), multiples, common factors, common multiples, highest common factor, lowest common multiple, prime factorisation, including using product notation and the unique factorisation theorem
- N5** apply systematic listing strategies
- N6** use positive integer powers and associated real roots (square, cube and higher), recognise powers of 2, 3, 4, 5
- N7** calculate with roots, and with integer indices
- N8** calculate exactly with fractions and multiples of π
- N9** calculate with and interpret standard form $A \times 10^n$, where $1 \leq A < 10$ and n is an integer

Fractions, decimals and percentages

What students need to learn:

- N10** work interchangeably with terminating decimals and their corresponding fractions (such as 3.5 and $\frac{7}{2}$ or 0.375 or $\frac{3}{8}$)
- N11** identify and work with fractions in ratio problems
- N12** interpret fractions and percentages as operators

Measures and accuracy

What students need to learn:

- N13** use standard units of mass, length, time, money and other measures (including standard compound measures) using decimal quantities where appropriate
- N14** estimate answers; check calculations using approximation and estimation, including answers obtained using technology
- N15** round numbers and measures to an appropriate degree of accuracy (e.g. to a specified number of decimal places or significant figures); use inequality notation to specify simple error intervals due to truncation or rounding
- N16** apply and interpret limits of accuracy